Screenshot #1

```
Last login: Sat Oct 17 23:07:26 on ttys001
hossamshribati@Mac ~ % cd ~/Tool-setup

cp -r lab04 lab05

cd lab05

ls -a
.                               uploader-app
..                              uploader-instance-id.txt
.env                            uploader-launch-data.json
create_app_stack_backup.sh      uploader-policy.json
create_app_stack.sh             user-data
delete_app_stack_backup.sh      viewer
delete_app_stack.sh             viewer-app
stack_outputs.env               viewer-instance-id.txt
testing                         viewer-launch-data.json
trust-policy.json               viewer-policy.json
uploader
hossamshribati@Mac lab05 %
```

Screenshot #2

```
================================== 1 passed in 1.53s ===================================
(.venv) hossamshribati@Mac lab05 % cd ~/Tool-setup/lab05

python -m pytest tests/ -v
================================== test session starts ==================================
platform darwin -- Python 3.13.1, pytest-9.1.1, pluggy-1.7.0 -- /Users/hossamshribati/To
ol-setup/lab05/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/hossamshribati/Tool-setup/lab05
collected 4 items

tests/test_aws_client.py::test_ec2_client PASSED                                  [ 25%]
tests/test_aws_client.py::test_s3_client PASSED                                   [ 50%]
tests/test_ec2_manager.py::test_create_instance PASSED                            [ 75%]
tests/test_s3_manager.py::test_create_bucket PASSED                               [100%]

================================== 4 passed in 1.99s ===================================
(.venv) hossamshribati@Mac lab05 %
```

Screenshot #3

```
tests/test_s3_manager.py::test_create_bucket PASSED                               [ 50%]
tests/test_s3_manager.py::test_delete_bucket PASSED                               [100%]

================================== 2 passed in 0.76s ===================================
(.venv) hossamshribati@Mac lab05 % cd ~/Tool-setup/lab05

python -m pytest tests/ -v
================================== test session starts ==================================
platform darwin -- Python 3.13.1, pytest-9.1.1, pluggy-1.7.0 -- /Users/hossamshribati/To
ol-setup/lab05/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/hossamshribati/Tool-setup/lab05
collected 6 items

tests/test_aws_client.py::test_ec2_client PASSED                                  [ 16%]
tests/test_aws_client.py::test_s3_client PASSED                                   [ 33%]
tests/test_ec2_manager.py::test_create_instance PASSED                            [ 50%]
tests/test_ec2_manager.py::test_delete_instance PASSED                            [ 66%]
tests/test_s3_manager.py::test_create_bucket PASSED                               [ 83%]
tests/test_s3_manager.py::test_delete_bucket PASSED                               [100%]

================================== 6 passed in 2.52s ===================================
(.venv) hossamshribati@Mac lab05 %
```

Screenshot #4

```
python -m pytest tests/ -v
================================= test session starts ==================================
platform darwin -- Python 3.13.1, pytest-9.1.1, pluggy-1.7.0 -- /Users/hossamshribati/To
ol-setup/lab05/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/hossamshribati/Tool-setup/lab05
collected 12 items

tests/test_alb_manager.py::test_create_load_balancer PASSED                      [  8%]
tests/test_alb_manager.py::test_delete_load_balancer PASSED                      [ 16%]
tests/test_asg_manager.py::test_create_auto_scaling_group PASSED                 [ 25%]
tests/test_asg_manager.py::test_delete_auto_scaling_group PASSED                 [ 33%]
tests/test_aws_client.py::test_ec2_client PASSED                                 [ 41%]
tests/test_aws_client.py::test_s3_client PASSED                                  [ 50%]
tests/test_aws_client.py::test_load_balancer_client PASSED                       [ 58%]
tests/test_aws_client.py::test_autoscaling_client PASSED                         [ 66%]
tests/test_ec2_manager.py::test_create_instance PASSED                           [ 75%]
tests/test_ec2_manager.py::test_delete_instance PASSED                           [ 83%]
tests/test_s3_manager.py::test_create_bucket PASSED                              [ 91%]
tests/test_s3_manager.py::test_delete_bucket PASSED                              [100%]

================================== 12 passed in 4.26s ==================================
(.venv) hossamshribati@Mac lab05 %
```

Screenshot #5

```
lab05 — root@0392935df869: /workspace — -zsh — 88×29

cachedir: .pytest_cache
rootdir: /Users/hossamshribati/Tool-setup/lab05
configfile: pytest.ini
collected 19 items

tests/unit/test_config.py::test_load_dotenv PASSED                               [  5%]
tests/unit/test_config.py::test_missing_required_setting PASSED                  [ 10%]
tests/unit/test_config.py::test_valid_settings PASSED                            [ 15%]
tests/unit/test_policies.py::test_trust_policy PASSED                            [ 21%]
tests/unit/test_policies.py::test_uploader_policy PASSED                         [ 26%]
tests/unit/test_policies.py::test_viewer_policy PASSED                           [ 31%]
tests/unit/test_stack_with_moto.py::test_create_bucket PASSED                    [ 36%]
tests/unit/test_stack_with_moto.py::test_create_role PASSED                      [ 42%]
tests/unit/test_stack_with_moto.py::test_create_instance_profile PASSED          [ 47%]
tests/unit/test_stack_with_moto.py::test_create_launch_template PASSED           [ 52%]
tests/unit/test_stack_with_moto.py::test_get_network PASSED                      [ 57%]
tests/unit/test_stack_with_moto.py::test_create_asg PASSED                       [ 63%]
tests/unit/test_stack_with_moto.py::test_create_load_balancer PASSED             [ 68%]
tests/unit/test_stack_with_moto.py::test_create_target_group PASSED              [ 73%]
tests/unit/test_stack_with_moto.py::test_create_listener PASSED                  [ 78%]
tests/unit/test_user_data.py::test_user_data_file_exists[uploader] PASSED        [ 84%]
tests/unit/test_user_data.py::test_user_data_file_exists[viewer] PASSED          [ 89%]
tests/unit/test_user_data.py::test_user_data_has_application_settings[uploader] PASSED [
 94%]
tests/unit/test_user_data.py::test_user_data_has_application_settings[viewer] PASSED [10
0%]

================================== 19 passed in 3.89s ==================================
(.venv) hossamshribati@Mac lab05 %
```

Screenshot #6

```
lab05 — root@0392935df869: /workspace — -zsh — 88×29

tests/unit/test_policies.py::test_trust_policy PASSED                    [ 33%]
tests/unit/test_policies.py::test_uploader_policy FAILED                 [ 66%]
tests/unit/test_policies.py::test_viewer_policy PASSED                   [100%]

=================================== FAILURES ===================================
_____________________________ test_uploader_policy _____________________________

    def test_uploader_policy():
        policy = json.loads(uploader_policy("test-bucket"))

        statement = policy["Statement"][0]

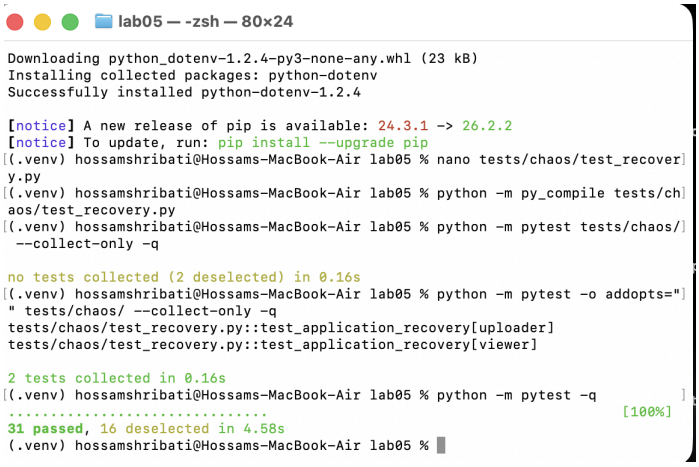
>       assert statement["Action"] == ["s3:DeleteObject"]
E       AssertionError: assert ['s3:PutObject'] == ['s3:DeleteObject']
E
E         At index 0 diff: 's3:PutObject' != 's3:DeleteObject'
E
E         Full diff:
E           [
E         -     's3:DeleteObject',
E         ?         ^^^^ -...
E
E         ...Full output truncated (3 lines hidden), use '-vv' to show

tests/unit/test_policies.py:27: AssertionError
=============================== short test summary info ================================
FAILED tests/unit/test_policies.py::test_uploader_policy - AssertionError: assert ['s3:P
utObject'] == ['s3:DeleteObject']
============================== 1 failed, 2 passed in 0.06s ==============================
```

Screenshot#7

```
 tests/unit/test_user_data.py::test_user_data_file_exists[uploader]
 tests/unit/test_user_data.py::test_user_data_file_exists[viewer]
 tests/unit/test_user_data.py::test_user_data_has_application_settings[uploader]
 tests/unit/test_user_data.py::test_user_data_has_application_settings[viewer]

 31/47 tests collected (16 deselected) in 0.39s
 (.venv) hossamshribati@Mac lab05 % cd ~/Tool-setup/lab05
[python -m pytest -q
 ................................                                         [100%]
 31 passed, 16 deselected in 7.05s
 (.venv) hossamshribati@Mac lab05 %
```

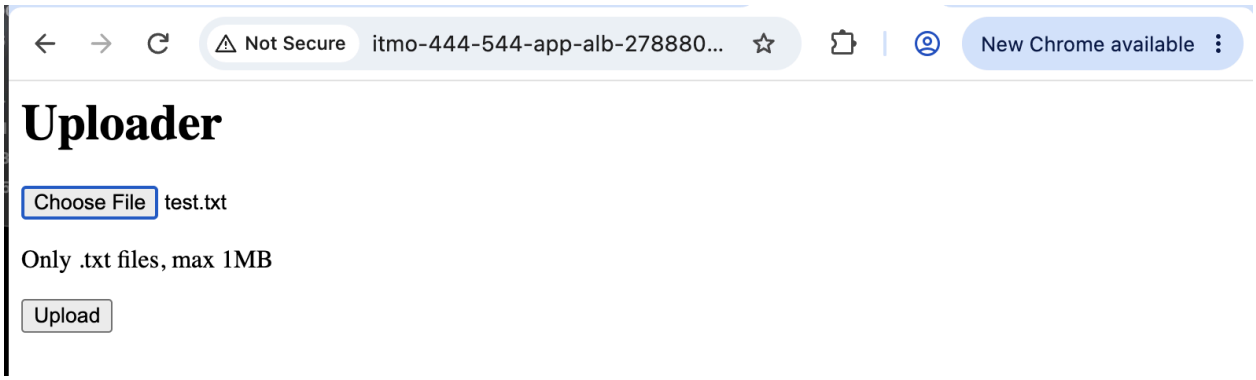


Websites URL

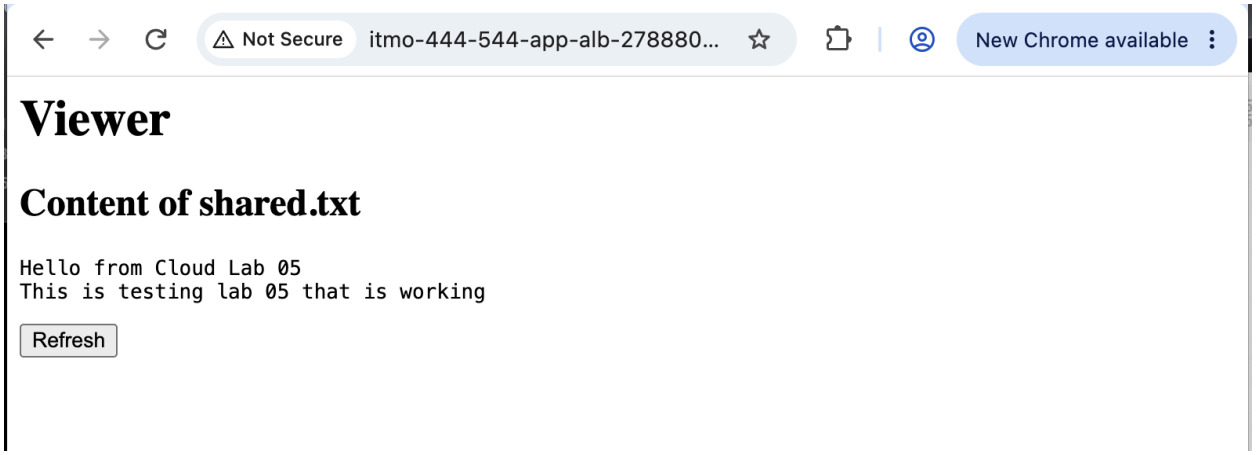
```
        )
(.venv) hossamshribati@Hossams-MacBook-Air lab05 % cd ~/Tool-setup/lab05 && nano stack/compute.py
(.venv) hossamshribati@Hossams-MacBook-Air lab05 % docker exec cloud-lab03 bash -c 'cd /workspace/lab05 && .venv-docker/bin/python -m py_compile stack/compute.py'
(.venv) hossamshribati@Hossams-MacBook-Air lab05 % docker exec cloud-lab03 bash -c 'cd /workspace/lab05 && .venv-docker/bin/python create_stack.py'
Stack created successfully!
Uploader URL: http://itmo-444-544-app-alb-278880882.us-east-2.elb.amazonaws.com:8080
Viewer URL: http://itmo-444-544-app-alb-278880882.us-east-2.elb.amazonaws.com:8081
(.venv) hossamshribati@Hossams-MacBook-Air lab05 %
```

Testing the websites





Testing viewer website




```
testpaths: tests
collecting ... collected 47 items / 33 deselected / 14 selected

tests/integration/test_aws_resources.py::test_load_balancer_exists PASSED [  7%]
tests/integration/test_aws_resources.py::test_target_groups_exist PASSED [ 14%]
tests/integration/test_aws_resources.py::test_uploader_auto_scaling_group_exists PAS
SED [ 21%]
tests/integration/test_aws_resources.py::test_viewer_auto_scaling_group_exists PASSE
D [ 28%]
tests/integration/test_aws_resources.py::test_uploader_iam_role_exists PASSED [ 35%]
tests/integration/test_aws_resources.py::test_viewer_iam_role_exists PASSED [ 42%]
tests/integration/test_aws_resources.py::test_uploader_launch_template_exists PASSED
 [ 50%]
tests/integration/test_aws_resources.py::test_viewer_launch_template_exists PASSED [
 57%]
tests/integration/test_stack_outputs.py::test_uploader_url_exists PASSED [ 64%]
tests/integration/test_stack_outputs.py::test_viewer_url_exists PASSED   [ 71%]
tests/integration/test_stack_outputs.py::test_alb_arn_exists PASSED      [ 78%]
tests/integration/test_stack_outputs.py::test_alb_dns_exists PASSED      [ 85%]
tests/integration/test_stack_outputs.py::test_uploader_target_group_exists PASSED [
92%]
tests/integration/test_stack_outputs.py::test_viewer_target_group_exists PASSED [100
%]

====================== 14 passed, 33 deselected in 3.51s =======================
(.venv) hossamshribati@Hossams-MacBook-Air lab05 %
```

Both EC2 instances recovered 180s

```
lab05 — -zsh — 84×26

tests/integration/test_stack_outputs.py::test_alb_dns_exists PASSED      [ 85%]
tests/integration/test_stack_outputs.py::test_uploader_target_group_exists PASSED [
92%]
tests/integration/test_stack_outputs.py::test_viewer_target_group_exists PASSED [100
%]

====================== 14 passed, 33 deselected in 3.13s =======================
(.venv) hossamshribati@Hossams-MacBook-Air lab05 % docker exec cloud-lab03 bash -c '
cd /workspace/lab05 && .venv-docker/bin/python -m pytest -o addopts="" -m chaos -s'
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.7.0
rootdir: /workspace/lab05
configfile: pytest.ini
testpaths: tests
collected 47 items / 45 deselected / 2 selected

tests/chaos/test_recovery.py
uploader: terminating i-0a4533ee3a1062302
uploader: recovery time = 133.72222155999043 seconds (SLO = 180 seconds)
.
viewer: terminating i-0180b75a18a67d4be
viewer: recovery time = 153.0281510700006 seconds (SLO = 180 seconds)
.

================= 2 passed, 45 deselected in 288.34s (0:04:48) =================
(.venv) hossamshribati@Hossams-MacBook-Air lab05 %
```

60s failed tests

```
        )

>       assert recovered_after is not None, (
            "The website did not recover in time"
        )
E       AssertionError: The website did not recover in time
E       assert None is not None

tests/chaos/test_recovery.py:83: AssertionError
========================== short test summary info ===========================
FAILED tests/chaos/test_recovery.py::test_application_recovery[uploader] - As...
FAILED tests/chaos/test_recovery.py::test_application_recovery[viewer] - Asse...
================= 2 failed, 45 deselected in 242.96s (0:04:02) =================
(.venv) hossamshribati@Hossams-MacBook-Air lab05 %
```

Stack_outputs.env output

```
''itmo-444-544-viewer-asg'\''}\n")'
(.venv) hossamshribati@Hossams-MacBook-Air lab05 % cd ~/Tool-setup/lab05 && cat stac
k_outputs.env
ALB_DNS=itmo-444-544-app-alb-278880882.us-east-2.elb.amazonaws.com
UPLOADER_TG_ARN=arn:aws:elasticloadbalancing:us-east-2:100995727963:targetgroup/itmo
-444-544-uploader-tg/af41bf4fe17af1d1
VIEWER_TG_ARN=arn:aws:elasticloadbalancing:us-east-2:100995727963:targetgroup/itmo-4
44-544-viewer-tg/2eb98aba85f4801f
UPLOADER_ASG_NAME=itmo-444-544-uploader-asg
VIEWER_ASG_NAME=itmo-444-544-viewer-asg
(.venv) hossamshribati@Hossams-MacBook-Air lab05 %
```

Stack deletion requested successfully!

```
(.venv) hossamshribati@Hossams-MacBook-Air lab05 % docker exec cloud-lab03 bash -c '
cd /workspace/lab05 && .venv-docker/bin/python delete_stack.py'
Deleted Auto Scaling Group: itmo-444-544-viewer-asg
Deleted Auto Scaling Group: itmo-444-544-uploader-asg
Stack deletion requested successfully!
(.venv) hossamshribati@Hossams-MacBook-Air lab05 %
```


```
(.venv) hossamshribati@Hossams-MacBook-Air lab05 % docker exec cloud-lab03 bash -c '
cd /workspace/lab05 && .venv-docker/bin/python delete_stack.py'
Deleted Auto Scaling Group: itmo-444-544-viewer-asg
Deleted Auto Scaling Group: itmo-444-544-uploader-asg
Stack deletion requested successfully!
(.venv) hossamshribati@Hossams-MacBook-Air lab05 %
```


No load balancers listed.

```
                                                ^
TARGET GROUPS:
bash: line 1: itmo-444-544-: command not found
(.venv) hossamshribati@Hossams-MacBook-Air lab05 % docker exec cloud-lab03 bash -c '
aws elbv2 describe-load-balancers --output table --no-cli-pager'
-----------------------
|DescribeLoadBalancers|
+---------------------+
(.venv) hossamshribati@Hossams-MacBook-Air lab05 %
```

Neither `itmo-444-544-uploader-tg` nor `itmo-444-544-viewer-tg` should appear.

```
(.venv) hossamshribati@Hossams-MacBook-Air lab05 % docker exec cloud-lab03 bash -c '
aws elbv2 describe-target-groups --output table --no-cli-pager'
----------------------
|DescribeTargetGroups|
+--------------------+
(.venv) hossamshribati@Hossams-MacBook-Air lab05 %
```

Auto Scaling Groups are gone.

```
(.venv) hossamshribati@Hossams-MacBook-Air lab05 % docker exec cloud-lab03 bash -c '
aws autoscaling describe-auto-scaling-groups --output table --no-cli-pager'
---------------------------
|DescribeAutoScalingGroups|
+-------------------------+
(.venv) hossamshribati@Hossams-MacBook-Air lab05 %
```

EC2 instances are also terminated

```
|||+-------------------------------------+------------------------------+|||
(.venv) hossamshribati@Hossams-MacBook-Air lab05 % docker exec cloud-lab03 bash -c
aws ec2 describe-launch-templates --output table --no-cli-pager'
-------------------------
|DescribeLaunchTemplates|
+-----------------------+
(.venv) hossamshribati@Hossams-MacBook-Air lab05 %
```

S3 bucket was deleted.

```
(.venv) hossamshribati@Hossams-MacBook-Air lab05 % docker exec cloud-lab03 bash -c 'aw
s s3api head-bucket --bucket cloud-lab05-files-100995727963'

aws: [ERROR]: An error occurred (404) when calling the HeadBucket operation: Not Found
(.venv) hossamshribati@Hossams-MacBook-Air lab05 %
```

IAM roles have been deleted

```
(.venv) hossamshribati@Hossams-MacBook-Air lab05 % docker exec cloud-lab03 bash -c 'a
ws iam get-role --role-name Lab05UploaderRole; aws iam get-role --role-name Lab05View
erRole'

aws: [ERROR]: An error occurred (NoSuchEntity) when calling the GetRole operation: Th
e role with name Lab05UploaderRole cannot be found.

aws: [ERROR]: An error occurred (NoSuchEntity) when calling the GetRole operation: Th
e role with name Lab05ViewerRole cannot be found.
(.venv) hossamshribati@Hossams-MacBook-Air lab05 %
```

instance profiles have also been deleted.

```
aws: [ERROR]: An error occurred (NoSuchEntity) when calling the GetInstanceProfile op
eration: Instance Profile Lab05UploaderProfile cannot be found.

aws: [ERROR]: An error occurred (NoSuchEntity) when calling the GetInstanceProfile op
eration: Instance Profile Lab05ViewerProfile cannot be found.
(.venv) hossamshribati@Hossams-MacBook-Air lab05 %
```

```
-------------------------------------------------
|            DescribeSecurityGroups             |
+----------------------+------------------------+
|  ITMO-444-544-lab-sg |  sg-0a74eba8fdadf9f88  |
+----------------------+------------------------+
(.venv) hossamshribati@Hossams-MacBook-Air lab05 % docker exec cloud-lab03 bash -c 'a
ws ec2 describe-instances --filters Name=instance-state-name,Values=running,pending,s
topping,stopped --query "Reservations[*].Instances[*].[InstanceId,State.Name]" --outp
ut table --no-cli-pager'
(.venv) hossamshribati@Hossams-MacBook-Air lab05 %
```

Written part

1. Moto helped me test my Python code without making real AWS resources. It showed that my code could create and check things like S3 buckets, IAM roles, security groups, and other AWS services in a test environment. However, Moto cannot make sure everything works correctly in real AWS. Because of this, I also used integration tests to test real AWS resources and chaos tests to see if the applications could recover after an EC2 instance was stopped or terminated.
2. **`ignore_errors()` vs. `2>/dev/null || true`** In Bash, `2>/dev/null || true` is shorter, but it hides all errors. This makes it hard to find problems because we cannot see why a command failed. In Python, `ignore_errors()` only ignores the AWS errors that I expect. Other errors are still shown. This makes the code safer and easier to fix when something goes wrong.
3. The Bash version was shorter because it used simple AWS CLI commands to create and delete AWS resources. The Python version needed more lines of code, but it made it easier to handle errors, reuse functions, and test the code with pytest. This makes the Python code easier to manage and fix when problems happen.